Dokumen Perancangan Perangkat Lunak



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FAKULTAS INFORMATIKA

PROGRAM STUDI INFORMATIKA

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(Dokumen Perancangan Perangkat Lunak)

Identifikasi Dokumen: DPPL-lMPAL-2026-01

Nama Sistem: SmartKasir (Point of Sale System)

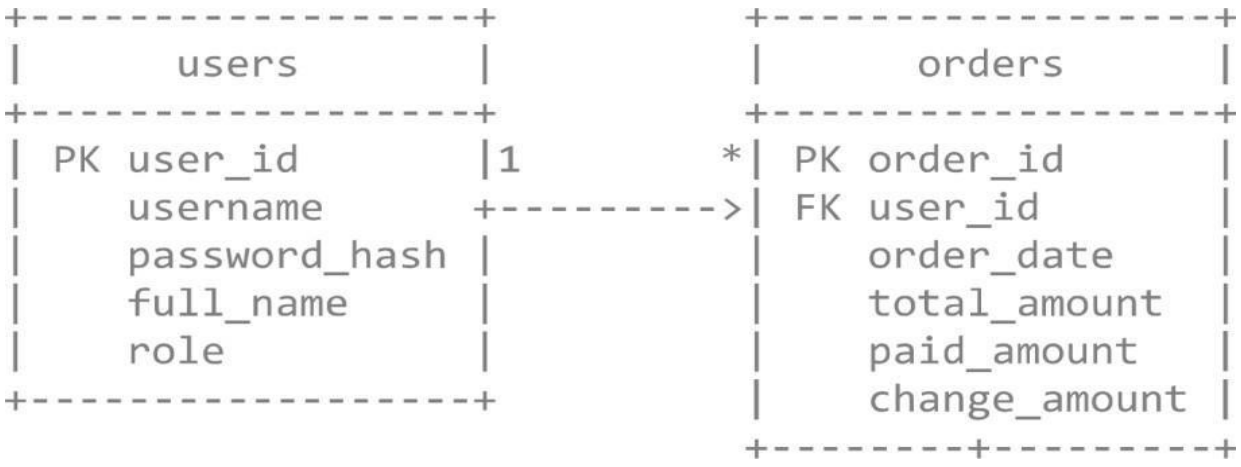
1. DESAIN ARSITEKTUR SISTEM

Sistem dirancang menggunakan arsitektur berlapis (Layered Pattern / MVC Pattern):

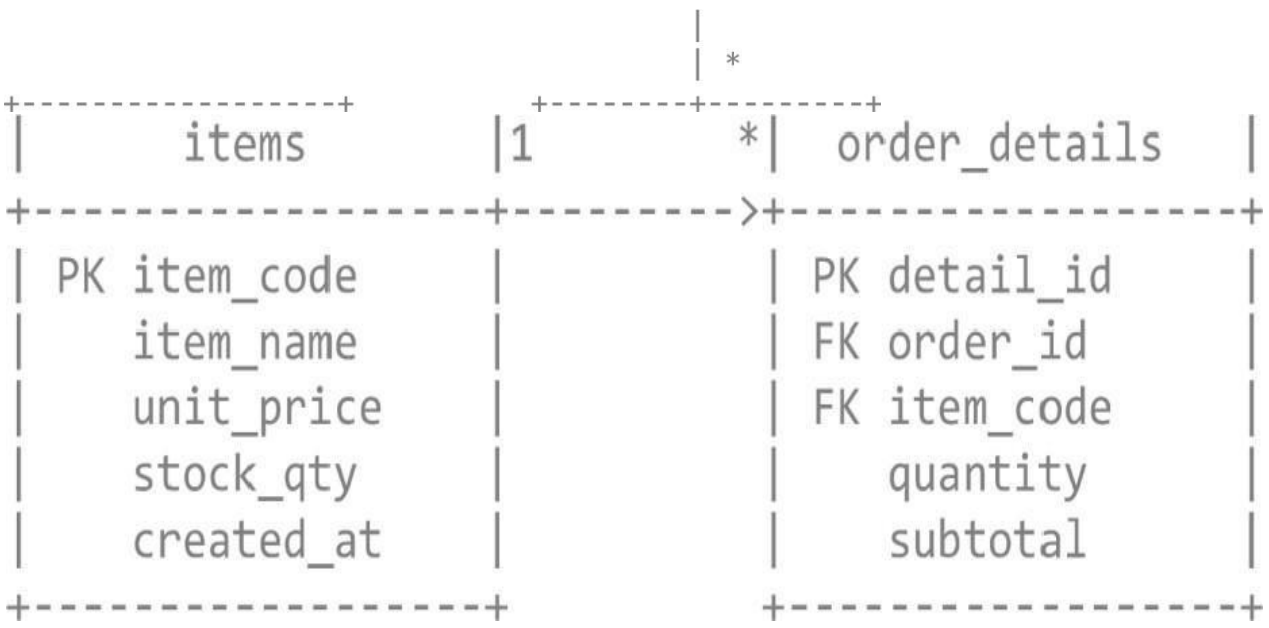
1. Presentation Layer (View): Menangani interaksi langsung dengan kasir/admin (Form Login, Layar Transaksi Kasir, Form Manajemen Barang).
2. Business Logic Layer (Controller / Service): Memvalidasi masukan kasir, aturan bisnis perhitungan diskon/pajak (jika ada), kalkulasi kembalian, orkestrasi pemotongan stok, dan service rendering struk belanja PDF.
3. Data Access Layer (Model / DAO): Bertanggung jawab atas kueri SQL transaksi (Data Definition & Manipulation) ke database engine (MySQL / PostgreSQL / SQLite).

2. DESAIN BASIS DATA (PHYSICAL DATA MODEL)

Skema Relasi Database



I 1



Kamus Data Tabel

Tabel: users

- user id (INT, Primary Key, Auto Increment)
- username (VARCHAR(50), Unique, Not Null)
- password_hash (VARCHAR(255), Not Null)
- full name (VARCHAR(IOO), Not Null)
- role (ENUM('ADMIN', 'CASHIER'), Not Null) Tabel: items

- item code (VARCHAR(30), Primary Key) • item name (VARCHAR(150), Not Null)

• unit_ price (DECIMAL(12,2), Not Null)

- stock_qty (INT, Not Null, Default: 0) created at (TIMESTAMP, Default: CURRENT_TIMESTAMP)

Tabel: orders

- order id (VARCHAR(35), Primary Key)— Format: TRXYYYYMMDD-XXXX • order date (DATETIME, Not Null) • user id (INT, Foreign Key users. user_id)
- total amount DECIMAL(12,2) Not Null) paid amount (DECIMAL(12,2), Not Null) change_amount (DECIMAL(12,2), Not Null)
- detail id (INT, Primary Key, Auto Increment) • order id (VARCHAR(35), Foreign Key + orders. order_id ON DELETE CASCADE) ○ item code (VARCHAR(30), Foreign Key + items . item_code) quantity (INT, Not Null)
- subtotal (DECIMAL(12,2), Not Null)

3. DESAIN KELAS DAN MODUL PROGRAM (CLASS DIAGRAM SPECIFICATION)

3.1 Entitas Model

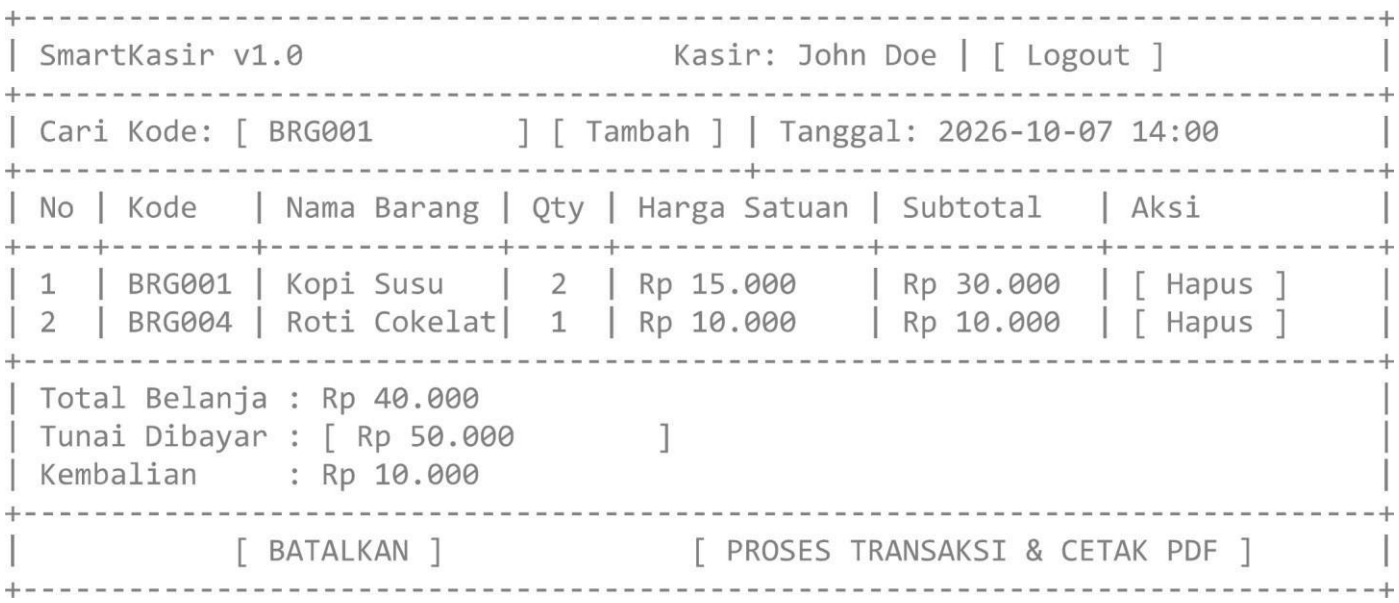
- User: Atribut (userId, username, passwordHash, role) .
- Item: Atribut (itemCode, itemName, unitPrice, stockQty).
- Order: Atribut (orderId, orderDate, userId, itemsList, totalAmount, paidAmount, changeAmount).
- OrderDetai1: Atribut (detailId, orderId, itemCode, quantity, subtotal).

3.2 Service & Controller

- AuthService:
- login(username, rawPassword) : UserSession o logout(): void o hashPassword(password) : String
- ProductService:
- get Item(itemCode) : Item o updateStock(itemCode, deltaQty) : boolean o saveProduct(Item) : boolean
- TransactionService:
- calculateSubtota1(unitPrice, qty): double o calculateGrandTota1(cartItems): double o validatePayment(grandTota1, paidAmount) : boolean
- calculateChange(grandTota1, paidAmount) : double
- commit Transaction (Order) : boolean (Menjalankan insert ke orders, order details, dan update items . stock_qty dalam satu transaksi database).
- PdfReceiptService:
- generateReceiptPdf(orderId) : File (Mengisi template struk: daftar item, total, nominal bayar, kembalian, nama kasir, dan waktu transak

4. PERANCANGAN ANTARMUKA PENGGUNA (UI WIREFRAME LAYOUT)

Layout 1: Layar Kasir / POS (Fokus Sprint 2)


SmartKasir v1.0
Kasir: John Doe | [Logout]
Cari Kode: [BRG001] [Tambah] | Tanggal: 2026-10-07 14:00
No | Kode | Nama Barang | Qty | Harga Satuan | Subtotal | Aksi
1 | BRG001 | Kopi Susu | 2 | Rp 15.000 | Rp 30.000 | [Hapus]
2 | BRG004 | Roti Cokelat | 1 | Rp 10.000 | Rp 10.000 | [Hapus]
Total Belanja : Rp 40.000
Tunai Dibayar : [Rp 50.000]
Kembalian : Rp 10.000
[BATALKAN]
[PROSES TRANSAKSI & CETAK PDF]